Enterprise Banking Intelligence & Early Warning Platform (EBIEWP)

Enterprise Data & Solution Architecture

---

1. Purpose

This document defines the logical data architecture and end-to-end solution architecture for the **Enterprise Banking Intelligence & Early Warning Platform (EBIEWP)**.

The architecture has been designed to support enterprise-scale Business Intelligence, customer analytics, lending analytics, customer experience analysis, Early Warning Intelligence and operational workflow automation.

The solution uses a **Star Schema** for the enterprise data warehouse and separates the platform into logical layers covering data generation, data engineering, analytical modelling, Business Intelligence, decision intelligence and operational automation.

The architecture is designed around the following principle:

Detect → Understand → Prioritize → Recommend → Route → Notify → Act

---

2. End-to-End Solution Architecture

```
         BUSINESS REQUIREMENTS
                  |
                  ▼
    ┌──────────────────────┐
    | Python Data Generation |
    | Synthetic Banking Data |
    └──────────┬───────────┘
               |
               ▼
    ┌──────────────────────┐
    |  Source Banking Data   |
    └──────────┬───────────┘
               |
               ▼
    ┌──────────────────────┐
    |   SQL Server ETL       |
    | Transform & Validate   |
    └──────────┬───────────┘
```

```
            │
            ▼
┌──────────────────────────────────────┐
│   ENTERPRISE DATA WAREHOUSE          │
│          STAR SCHEMA                 │
│                                      │
│ Dimensions        Facts              │
│ ──────────        ─────              │
│ DimCustomer       FactTransaction    │
│ DimBranch         FactLoan           │
│ DimAccount        FactRepayment      │
│ DimProduct        FactComplaint      │
│ DimDate           FactDigital...     │
└──────────────────┬───────────────────┘
            │
            ▼
┌──────────────────────────┐
│ SQL Analytical Layer     │
│ Views & Procedures       │
└────────────┬─────────────┘
            │
            ▼
┌──────────────────────────┐
│ Power BI Semantic Model   │
│ SQL + DAX                │
└────────────┬─────────────┘
            │
┌────────────┴─────────────┐
▼            ▼
┌──────────────────┐  ┌──────────────────────┐
│ Banking Analytics│  │ Customer 360         │
│ & Dashboards     │  │ Intelligence         │
└────────┬─────────┘  └──────────┬───────────┘
         │                       │
         └───────────┬───────────┘
            ▼
┌──────────────────────────┐
│ Early Warning            │
│ Intelligence             │
└────────────┬─────────────┘
            │
            ▼
┌──────────────────────────┐
│ Risk Prioritization      │
│ & Recommended Action     │
└────────────┬─────────────┘
            │
            ▼
```

```
┌──────────────────────────┐
│ Stakeholder Routing      │
│ + Stakeholder Email      │
└────────────┬─────────────┘
             │
             ▼
┌──────────────────────────┐
│ n8n Automation Workflow  │
└────────────┬─────────────┘
             │
             ▼
┌──────────────────────────┐
│ Automated Email Alert    │
│ via Gmail                │
└────────────┬─────────────┘
             │
             ▼
┌──────────────────────────┐
│ Responsible Stakeholder  │
│ & Business Intervention  │
└──────────────────────────┘
```

---

3. Architecture Layers

The platform consists of the following logical layers:

1. Business Requirements Layer
2. Data Generation Layer
3. Source Data Layer
4. Data Engineering / ETL Layer
5. Enterprise Data Warehouse Layer
6. Analytical / Semantic Layer
7. Business Intelligence Layer
8. Customer 360 Intelligence Layer
9. Early Warning Intelligence Layer
10. Decision & Prioritization Layer
11. Stakeholder Routing Layer
12. Workflow Automation Layer
13. Business Intervention Layer

---

4. Data Generation Layer

Python is used to generate realistic synthetic banking datasets.

The purpose of the generation layer is to provide a controlled environment in which enterprise banking data engineering, analytics and decision-intelligence concepts can be demonstrated without using real customer information.

The generated data covers areas including:

- Customers
- Accounts
- Branches
- Products
- Transactions
- Loans
- Loan repayments
- Complaints
- Digital activity
- Calendar dates

The generated datasets are designed to support realistic relationships and business behaviour across the banking domains.

---

5. Source Data Layer

The source layer represents operational banking information before it is transformed into the analytical warehouse.

The major business domains include:

- Customer management
- Account management
- Banking transactions
- Lending
- Loan repayment
- Customer complaints
- Digital activity
- Branch operations

These source datasets provide the foundation for downstream analytics.

6. Data Engineering & ETL Layer

The SQL Server ETL layer transforms source datasets into structured, validated warehouse data.

The ETL process includes:

- Data ingestion
- Data transformation
- Data cleansing
- Business-rule enforcement
- Dimension loading
- Fact loading
- Referential integrity validation
- Duplicate validation
- Mandatory-field validation
- Row-count validation
- Warehouse health checks

The objective is to ensure that downstream BI processes consume trusted and analytically consistent data.

7. Enterprise Data Warehouse

The Enterprise Data Warehouse uses a **Star Schema** approach.

The warehouse separates descriptive business entities into dimensions and measurable business events into fact tables.

7.1 Dimension Tables

Dimension	Purpose
DimCustomer	Customer demographics, segmentation and banking profile
DimBranch	Branch hierarchy and geographic information
DimAccount	Customer account portfolio
DimProduct	Banking product information
DimDate	Enterprise calendar dimension

7.2 Fact Tables

Fact	Purpose
FactTransaction	Banking transaction history
FactLoan	Lending portfolio and loan information
FactLoanRepayment	Loan repayment activity and performance
FactComplaint	Customer complaint and service activity
FactDigitalActivity	Digital banking activity

---

8. Logical Star Schema

```
                DimDate
                   │
                   │
                   ▼
DimCustomer ───────── FactTransaction ───────── DimAccount
    │
    │
    ├──────────────── FactLoan ──────────────── DimProduct
    │                    │
    │                    │
    │                    ▼
    │          FactLoanRepayment
    │
    └──────────────── FactComplaint

DimBranch
   │
   ├────────────── FactTransaction
   ├────────────── FactLoan
   ├────────────── FactComplaint
   └────────────── FactDigitalActivity

DimProduct
   │
   └────────────── FactLoan
```

The exact physical relationships and foreign-key implementation are maintained in the SQL Server database scripts.

---

9. SQL Analytical Layer

The SQL analytical layer provides business-ready structures above the physical warehouse.

This layer includes:

- Analytical SQL views
- Stored procedures
- Business-rule calculations
- Aggregated analytical structures
- Warehouse validation procedures

The purpose of the layer is to simplify downstream analytical consumption and centralize reusable business logic where appropriate.

---

10. Power BI Semantic Layer

Power BI consumes the trusted analytical data and provides the business-facing semantic layer.

The semantic model incorporates:

- SQL Server data
- DAX measures
- DAX calculated columns
- DAX calculated tables
- Business rules
- Customer health logic
- Early Warning logic
- Recommended action logic
- Stakeholder routing logic

The semantic layer translates technical warehouse data into business concepts that can be understood and used by decision-makers.

---

11. Business Intelligence Layer

The Power BI reporting layer provides executive and operational visibility across major banking domains.

The completed dashboard solution covers:

- Executive Banking Overview
- Customer 360 / Customer Intelligence
- Loan Portfolio
- Credit Risk
- Branch Performance
- Complaint Intelligence
- Early Warning Intelligence

The dashboards use interactive filtering and business-focused visualizations to support investigation from executive-level indicators down to customer-level cases.

---

12. Customer 360 Intelligence Layer

Customer 360 consolidates information across financial, credit and service domains.

The Customer 360 intelligence layer includes indicators such as:

- Customer profile
- Customer segment
- Branch
- Region
- Account relationships
- Deposit activity
- Transaction activity
- Loan exposure
- Outstanding principal
- Payment behaviour
- Days Past Due
- Consecutive missed payments
- Complaint volume
- Repeat complaints
- Unsatisfied complaints
- Customer health

This layer provides the foundation for customer-level Early Warning Intelligence.

---

13. Early Warning Intelligence Layer

The Early Warning layer converts customer-level business signals into explainable warning classifications.

The framework evaluates financial, credit and service indicators and produces:

- Customer Health
- Warning Reason
- Warning Severity
- Warning Type
- Recommended Action
- Stakeholder
- Stakeholder Email
- Financial Exposure

Example warning classifications include:

- Credit Deterioration
- Service Deterioration
- Credit + Service

Customer health categories include:

- Healthy
- Watchlist
- At Risk

Severity categories include:

- High
- Medium

---

14. Decision & Prioritization Layer

The platform prioritizes customers requiring attention using explainable business rules.

Priority is determined using:

1. Warning severity
2. Days Past Due
3. Complaint volume
4. Outstanding principal

This ensures that stakeholders can focus on the cases with the greatest urgency and potential business impact.

The decision layer also determines the recommended intervention.

Examples include:

- Relationship Manager Follow-up
- Customer Service Follow-up
- Service Recovery

---

15. Stakeholder Routing Layer

After an intervention is recommended, the platform determines the responsible stakeholder.

The routing model includes:

Recommended Action	Stakeholder
Relationship Manager Follow-up	Branch Manager
Customer Service Follow-up	Customer Service
Service Recovery	Service Recovery Team

Stakeholder email information is included in the Early Warning Automation dataset so that the automation workflow can route alerts to the appropriate recipient.

For the demonstration environment, stakeholder email addresses are simulated rather than connected to real bank identities.

---

16. Early Warning Automation Dataset

A dedicated **Early Warning Automation** dataset is generated from the Customer 360 intelligence layer.

The dataset contains the fields required to support operational alerting, including:

- CustomerID
- CustomerName
- BranchID
- BranchName
- Region
- CustomerHealth

- DPD
- ConsecutiveMissedPayments
- RepeatComplaints
- TotalComplaints
- UnsatisfiedComplaints
- PaymentBehaviour
- WarningReason
- WarningSeverity
- WarningType
- RecommendedAction
- TotalOutstandingPrincipal
- Stakeholder
- StakeholderEmail

This dataset acts as the interface between Power BI intelligence and the workflow automation layer.

---

17. Workflow Automation Layer

The automation layer uses **n8n** to transform Early Warning Intelligence into stakeholder notifications.

The workflow is:

```
Early Warning Automation Dataset
        |
        ▼
   CSV Export
        |
        ▼
  Google Drive
        |
        ▼
      n8n
        |
        ▼
 CSV Record Processing
        |
        ▼
Stakeholder Grouping
        |
        ▼
HTML Email Generation
```

|
▼
Gmail
|
▼
Stakeholder Notification

---

18. n8n Platform Selection

Power Automate was considered because EBIEWP is built primarily around Microsoft technologies, particularly SQL Server and Power BI.

However, n8n was selected for the demonstration because the project was developed in a personal environment rather than an organizational Microsoft work or school tenant.

n8n provided a practical way to demonstrate workflow orchestration without making the portfolio implementation dependent on organizational Microsoft account, tenant and licensing arrangements.

This is a demonstration-environment decision rather than a claim that n8n is inherently superior to Power Automate.

In a production organization already standardized on Microsoft 365, Power Automate would be a strong alternative because of its native integration with Microsoft services.

The architectural principle remains the same regardless of orchestration platform:

BI Insight → Business Decision → Automated Workflow → Stakeholder Action

---

19. Automated Email Architecture

The email generation process creates stakeholder-specific HTML notifications.

Each notification can contain:

- Stakeholder name
- Total alert count
- High-severity alerts
- Medium-severity alerts
- At-Risk customers
- Watchlist customers
- Total outstanding principal

- Leading warning drivers
- Recommended action
- Priority customer table

Priority customers are ranked using:

1. Warning severity
2. Days Past Due
3. Complaint volume
4. Outstanding principal

This allows stakeholders to receive a concise operational summary rather than raw customer-level data.

---

20. End-to-End Business Process

The completed EBIEWP process can be summarized as:

OPERATIONAL DATA
↓
DATA ENGINEERING
↓
ENTERPRISE DATA WAREHOUSE
↓
ANALYTICS
↓
POWER BI
↓
CUSTOMER 360
↓
EARLY WARNING
↓
RISK PRIORITIZATION
↓
RECOMMENDED ACTION
↓
STAKEHOLDER ROUTING
↓
n8n AUTOMATION
↓
EMAIL NOTIFICATION
↓
BUSINESS INTERVENTION

This architecture demonstrates the transition from traditional reporting to operational decision intelligence.

---

21. Data Quality & Governance

Data quality is addressed throughout the platform.

Validation includes:

- Dataset validation
- Row-count validation
- Duplicate detection
- Mandatory-field validation
- Referential integrity
- Business-rule validation
- Analytical layer validation
- Stored procedure validation
- Warehouse health validation

The platform is designed so that analytical decisions are based on structured and validated information.

---

22. Security & Production Considerations

EBIEWP is a portfolio demonstration using synthetic data.

A production implementation would require:

- Enterprise identity and access management
- Role-based security
- Secure credential management
- Secrets management
- Data encryption
- Audit logging
- Data lineage
- Persistent alert state
- Alert deduplication
- Intervention history
- SLA monitoring

- Automation monitoring
- Failure handling
- Production case-management integration

These capabilities are outside the scope of the portfolio demonstration.

---

23. Architecture Decision Summary

Decision	Rationale
Python for data generation	Reproducible synthetic banking data generation
SQL Server for warehouse	Structured enterprise relational storage and analytical processing
Star Schema	Efficient dimensional analytics and Power BI consumption
SQL Views / Procedures	Reusable analytical and business logic layer
Power BI	Executive and operational Business Intelligence
DAX	Semantic calculations and business intelligence logic
Customer 360	Unified customer-level intelligence
Rules-based Early Warning	Explainable and transparent risk detection
n8n	Flexible automation in personal demonstration environment
Google Drive	Practical workflow file exchange
Gmail	Demonstration stakeholder notification

---

24. Final Architecture Principle

The architecture is intentionally designed to connect technology layers to a business outcome.

It does not end with a dashboard.

It follows the complete decision chain:

DATA
↓
INFORMATION
↓

INSIGHT
↓
WARNING
↓
PRIORITY
↓
DECISION
↓
STAKEHOLDER
↓
ACTION

This is the core architectural principle of the Enterprise Banking Intelligence & Early Warning Platform.

---

25. Project Scale

Component	Volume
Customers	100,000
Accounts	153,035
Branches	150
Calendar Dates	3,834
Transactions	100M+
Active Loans	4,937
Repayment Records	33,000+
Complaint Records	1.13M+
Early Warning Records	47,847
Stakeholder Groups	13
Demonstration Stakeholder Emails	13

---

26. Architecture Status

Architecture Component	Status
Business Requirements	Complete
Python Data Generation	Complete
Enterprise Data Warehouse	Complete
SQL Server ETL	Complete
SQL Analytical Layer	Complete

Architecture Component	Status
Power BI Semantic Model	Complete
Executive & Operational Dashboards	Complete
Customer 360 Intelligence	Complete
Early Warning Intelligence	Complete
Stakeholder Routing	Complete
n8n Automation	Complete
Automated Email Demonstration	Complete

---

27. Conclusion

The EBIEWP architecture provides an end-to-end framework for transforming synthetic operational banking data into actionable business intelligence.

The architecture combines:

Enterprise Data Engineering

with

Business Intelligence

with

Customer 360 Intelligence

with

Early Warning Decision Intelligence

with

Workflow Automation

to create a complete analytical-to-operational business process.

The final solution demonstrates:

> **Detect → Understand → Prioritize → Recommend → Route → Notify → Act**